崑山科技大學日間部先進應用材料工程系大學部四技綠能科技技優領航專班課程科目表

Kun Shan University Department of Advanced Applied Materials Engineering

4-Year Green energy technology technical class College Curricula

課程名稱(講授時數－實習時數－學分數)

Course Name (Lecture Hours – Practice Hours – Credits)

111.09.29 系課程委員會通過 Approved by the Department Curriculum Committee
111.11.14 院課程委員會通過 Approved by the College Curriculum Committee
112.02.13 校課程委員會修正通過 Approved by the University Curriculum Committee
112 學年度入學適用
Effective for students admitted in Fall, 2024

第一學年 (1st Academic Year)		第二學年 (2st Academic Year)		第三學年 (3st Academic Year)		第四學年 (4st Academic Year)		
第一學期 1st Semester	第二學期 2st Semester	第一學期 1st Semester	第二學期 2st Semester	第一學期 1st Semester	第二學期 2st Semester	第一學期 1st Semester	第二學期 2st Semester	暑期 Summer
共同必修科目(合計 30 學分) General Courses (Total：30 Credits)								
體育分組(一) Physical Education (I) (2-0-2) 中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2) 英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	體育分組(二) Physical Education (II) (2-0-2) 勞作教育(一) Labor Education (I) (0-1-0.5) 中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2) 英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2) 校訂分類通識 School-designated Liberal Education (2-0-2)	勞作教育(二) Labor Education (II) (0-1-0.5) 校訂分類通識 School-designated Liberal Education (2-0-2) 院訂核心通識 College-designated General Core **** (2-0-2) 院訂分類通識 College-designated Liberal Education ***** (2-0-2)	服務學習與公民教育 Public Service and Civil Education (0-1-0) 校訂分類通識 School-designated Liberal Education (2-0-2)	生涯發展與職業倫理 Career Development and Work Ethics (2-0-2) 院訂分類通識 College-designated Liberal Education ***** (2-0-2)	校訂核心通識 -外語選讀*** School-Foreign Languages *** (2-0-2)			英文能力檢定 English Proficiency Examination* (1-0-1)
6-0-6	8-1-8.5	6-1-6.5	2-1-2	4-0-4	2-0-2			(1-0-1)
共同選修科目(合計 1 學分) Common Elective (Total：1 Credits)								
								跨領域學習護照 Interdisciplinary Learning Passport (1-0-1)
								(1-0-1)

註：Note

* 「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。 “English Proficiency Examination” is enacted according to “Kun Shan University Guidelines for English Proficiency Requirement.”

** 跨領域學習護照，依「崑山科技大學通識學習護照實施辦法」辦理。 “Interdisciplinary Learning Passport” is enacted according to “Kung Shan University Guidelines for General Education Learning Passport.”

*** 「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開 設於三年級，然「第二外語」課程為二年級優先選課。 “Foreign Languages” compromises “Foreign Languages (other than English)” and “Workplace English”, among which students choose 1 course. While the courses are offered in the third academic year, “Foreign Languages (other than English)” are given preference to the sophomores.

**** 「院訂核心通識」，以開設英文相關課程為原則，課程科目表上開課時間為暫訂，課程名稱內容由各 院自行規劃開設 English learning related content is given preference to offer in principle for college-designated general core courses. The schedule of this course on this curriculum chart is tentative. Course names and content are developed by the individual college.

***** 「院訂分類通識」，分別為一門數位科學類及一門院規劃通識，課程科目表上開課時間為暫訂，課程 名稱內容由各院自行規劃開設 College-designated liberal education comprises one course in digital science, and one course developed by the individual colleges. The schedule of these courses on this curriculum chart are tentative. Course names and content are developed by the individual college.

專業必修科目(合計 29 學分) Compulsory Courses (Total: 29 Credits)

材料科學概論(一) Introduction of Materials Science(I) (3-0-3)	金屬材料 Metallic Materials (3-0-3)	物理冶金 Physical Metallurgy (3-0-3) 基礎儀器分析 Instrumental Analysis (2-2-4)	陶瓷材料 Ceramic Materials (3-0-3)	材料加工實驗 Materials Processing and Experiments (0-3-3) 實務專題(一) Practice and Study on Selected Topic (I) (0-1-1)	材料檢測與分析實驗 Materials Testing and Analysis Experiments (0-3-3) 實務專題(二) Practice and Study on Selected Topic (II) (0-1-1)	學術倫理* Academic Ethics (1-0-1)	材料研討課程 Seminar For Materials (0-2-2)	暑期校外實習(一)** Summer out of school internship(I) (0-*-2)
3-0-3	3-0-3	5-2-7	3-0-3	0-4-4	0-4-4	1-0-1	0-2-4	

註：Note:

1.*「學術倫理」之授課與學分取得依課務組網頁之「學術倫理教育實施辦法」辦理。
The implementation of the "Academic Ethics" course and the acquisition of its credits are carried out in accordance with the Regulations for Academic Ethics Education Implementation on the Academic Affairs Division website.

2.**暑期校外實習課程依材料系之「校外實習必修課程作業要點」辦理。
The summer off-campus internship course is carried out in accordance with the Implementation Guidelines for Off-campus Internship Courses of the Department of Advanced Applied Materials Engineering.

3.***「專業證照」課程依材料系之「專業領域基本能力檢定實施辦法」辦理
Professional certification courses are carried out in accordance with the Implementation Guidelines for Basic Professional Competency Assessments of the Department of Advanced Applied Materials Engineering.

專業選修科目(33 學分) (每學期依實際情況開課) Elective Courses(Total: 33 Credits) (Based on every semester opening courses)

工程技術概論 Introduction to Engineering Technology (2-0-2) 微積分 Calculus (2-0-2) 普通化學(一) General Chemistry(I) (3-0-3) 普通物理(一) University Physics(I) (2-0-2) 普通化學實驗 General Chemistry Experiments (0-2-2) 工業工程概論 Introduction of Industrial Engineering (3-0-3) 綠色材料 Green Materials (3-0-3)	工程數學 Engineering Mathmatics (2-0-2) 普通化學(二) General Chemistry(II) (3-0-3) 普通物理(二) University Physics(II) (2-0-2) 普通物理實驗 Physics Experiments (0-2-2) 材料科學概論(二) Introduction of Materials Science(II) (3-0-3) 工程塑膠 Engineering Plastics (2-0-2)	材料力學 Mechanics of Materials (3-0-3) 基礎儀器分析 Instrumental Analysis (1-2-3) 計算機程式設計 Computer Programming (1-0-1) 有機化學 Organic Chemistry (3-0-3)	材料熱力學 Thermodynamics of Materials (3-0-3) 材料製備實驗 Materials Preparation Experiments (0-3-3) 材料加工 Materials Processing (2-0-2) 專業英文 English for Specific Purposes (2-0-2) 電化學 Electrochemical (3-0-3) 光電材料 Optoelectronic Materials (3-0-3)	材料檢測與分析 Materials Testing and Analysis (2-0-2) 高分子科學 Polymer Science (3-0-3) 複合材料 Composite Materials (3-0-3)	功能性材料 Functional Materials (3-0-3) 職業安全與衛生 Occupational Safety and Health (3-0-3)	學年校外實習(一) Academic year Internship (I) (0-*-9)** 工廠管理 Industry Management (2-0-2)	創意與創業 Creativity and Entrepreneurship (3-0-3) 學年校外實習(二) Academic year Internship (II) (0-*-9)** 學期校外實習 Semester Intership (0-*-9)**	暑期校外實習(二) Summer out of school internship(II) (0-*-2)**

註：Note:

**暑期校外實習(一)(二)、學期校外實習、學年校外實習(一)(二)等課程依「崑山科技大學學生校外實習辦法」辦理
The Summer Off-campus Internship (I)(II), Semester Internship, and Academic Year Internship (I)(II) courses are conducted in accordance with the Regulations for Student Off-campus Internships of Kun Shan University.

技優生推薦選修之三創與跨領域課程Recommended electives for gifted students: three innovative and cross-field courses

※三創選修課程(須修至少 18 學分) Three elective courses (required to take at least 18 credits)

創意程式設計-C-E Creative programming -C-E (2-0-2) 創意媒體與生活 Creative media and life (1-0-1)	創意故事 creative story (2-0-2) 創意管理 creative management (3-0-3)	創意電子控制工具使用入門-E Getting Started with Creative Electronic Control Tools-E (2-0-2) 創意媒體英文表達-院核心 Creative Media English ExpressionHouse Core (2-0-2) 創意文案撰寫 Creative Copywriting (1-2-3)	創意行銷 Creative Marketing (2-0-2) 創意開發 Creative Development (2-0-2) 電腦輔助創新設計 ComputerAided Innovative Design (1-2-3)	綠色科技與創新-C Green Technology and Innovation-C (2-0-2) 創意思考與方法 Creative Thinking and Methods (2-0-2) 廣告創意與文案 Advertising Creativity and Copywriting (1-1-2)	創新事業導論-B Introduction to Innovation Business-B (2-0-2) 創新智慧生活科技 Innovative Smart Life Technology (2-1-3)	創意思考訓練 Creative Thinking Training (2-0-2) 創意與專利 Creativity and Patent (3-0-3) 創意與創業 Creativity and Entrepreneurship (3-0-3)	就業與創業技能輔導 Employment and Entrepreneurship Skills Counseling (3-0-3) 環保與創意 Environmental Protection and Creativity (3-0-3) 創業與經營實務 Entrepreneurship and Business Practice (2-1-3)

※跨領域選修課程 Interdisciplinary elective courses (at least 18 credits required)

綠能科技與實作 Green Energy Technology and Implementat ion (2-1-3) 太陽能工程概論 Introduction to Solar Engineering (3-0-3) 燃料電池應技術 Fuel Cell Technology (3-0-3) 電子商務 ECommerce (3-0-3)	工程材料學 Engineering Materials (3-0-3) 工程技術概論 Introduction to Engineering Technology (2-0-2) 遊戲企劃 Game Planning （3- 0-3）	智慧車輛控制系統 Intelligent Vehicle Control System (3-0-3) 智慧製造技術 Smart Manufacturi ng Technology (3-0-3) 多媒體製作 Multimedia Production (3-0-3) 遊戲行為分析 Game Behavior Analysis (2-0-2)	物聯網與大數據於智慧製造應用 Internet of Things and Big Data in Smart Manufacturin g Applications (3-0-3) 工業 4.0 特色技術 Industry 4.0 Featured Technologies (3-0-3) 智慧商務應用 Smart Business Applications (3-0-3) 智慧商圈管理 Smart Business District Management (3-0-3)	新能源與人文社會社會發展 New Energy and Humanistic and Social Development (2-0-2) 綠色能源概論 Introduction to Green Energy (3-0-3) 材料科學導論 Introduction to Materials Science (3-0-3) 綠建築與智慧建築 Green Building and Smart Building (3-0-3) 行動遊戲製作 Mobile Game Production (2-1-3)	儲能系統概論 Introduction to Energy Storage Systems (3-0-3) 綠色能源與產業商機 Green Energy and Industrial Business Opportunitie s (3-0-3) 企業節能減碳概論 Introduction to Enterprise Energy Saving and Carbon Reduction (3-0-3) 環境規劃與管理 Environmental Planning and Management (3-0-3) 能源應用 Energy Applications (3-0-3) 遊戲行銷營運實務 Game Marketing Operation Practice (0-2-2)	環工生質能源 Environment al Biomass Energy (3-0-3) 溫室氣體管理 Greenhouse gas Management (3-0-3) 光電材料 Optoelectron ic Materials (3-0-3) 節能材料概論 Introduction to EnergySaving Materials (3-0-3) 太陽光電能發電系統 Solar Photoelectric Power Generation System (3-0-3)	物聯網於綠能之應用 Application of Internet of Things in Green Energy (3-0-3) 智慧能源與物聯網 Smart Energy and Internet of Things (3-0-3) 物聯網與綠能概論 Introduction to Internet of Things and Green Energy (2-0-2) 物聯網與綠能應用概論 Introduction to Internet of Things and Green Energy Applications (3-0-3) 物聯網與綠能應用實務 Internet of Things and Green Energy Application Practice (3- 0-3)

最低畢業學分：128 學分 Minimum for Total Credits: 128 Credits

註：

1. 本系最低畢業學分為 128 學分，必修 59 學分(包含共同科目、系專業必修科目)、選修 69 學分(系專業選修科目、三創選修課程、跨領域選修課程及學生自行選修經系/學位學程認可之專業課程)。

The minimum graduation credits of this department are 128 credits: 65 credits of compulsory courses and 63 credits of elective courses

2. 跨域修課規定：本系日間部大學部學生除修讀本系專業課程外，另需修讀外系（不限學院）專業課程至少 9 學分(不含學術倫理及單一科目大於等於 5 學分之特殊課程)，以強化學生跨領域知識，增加就業力。

Interdisciplinary Course Requirements: In addition to their own departmental courses, daytime students of this department must complete at least 9 credits of courses from other departments (regardless of college) to enhance their interdisciplinary knowledge and improve their employability.(Excluding academic ethics courses and special courses with a single subject of 5 or more credits.)

3. 暑期校外實習(必修)實習時數至少 160 小時；學期校外實習 (選修)開設共 9 學分為連續 4.5 個月之實習，學年校外實(一)(二) (選修)開設共 18 學分為連續 9 個月之實習 ，相關辦法另訂之。

The Summer Off-campus Internship (compulsory) requires a minimum of 160 hours. The Semester Internship (elective) offers 9 credits and consists of a continuous internship period of 4.5 months. The Academic Year Internship (I)(II) (elective) offers 18 credits and

consists of a continuous internship period of 9 months. Relevant regulations are stipulated separately.

4. 本技優專班課程，規劃 18 學分以上的三創課程及跨領域等選修課程，協助技優生運用優秀的技術能力及創造力，呈現多元的創作能量，體現技職教育的特色。

This special class for technical students plans more than 18 credits of three innovative courses and cross-field elective courses to help technical students use their excellent technical abilities and creativity to present diverse creative energy and reflect the characteristics of technical and vocational education

5. 學生須修畢本系材料應用數位科技微學程課程始得畢業。

Students in this department must complete the Micro Program in Digital Technology for Materials Applications and obtain the corresponding certificate before graduation.

6. 學生須修畢本院跨院系學程或本校他院之跨院系學程，始得畢業。

Students are required to complete at least one cross-college interdisciplinary program offered by the university in order to graduate.